

**ADDIS ABABA UNIVERSITY,
FACULTY OF SCIENCE**



Department of Mathematics and Computer Science

**Entrance examination for the Masters of Science
Programme in Computer Science**

Date : June 4, 2002
Time allowed: 3 hours

Full Name : _____

Grade Obtained : _____

Important Note

- Answer neatly to all the questions on the space provided on the question paper
- Write short, and grammatically and logically correct sentences

Part I: Computer organization and assembly (11 pts total)

1. Base your answer to the questions below on the following fragment of code that sorts a list of numbers in ascending order using the bubble sort algorithm. Assume that the list is already stored in the memory.

```
        mov cx, p
        dec cx
lp1:    mov di, cx
        mov bx, 0
lp2:    mov ax, a[bx]
        cmp ax, a[bx+2]
        jle continue
        xchg ax, a[bx+2]
        mov a[bx], ax
continue: add bx, 2
        loop lp2
        mov cx, di
        loop lp1
```

1.1. (1 pt) List the variables used in the code.

1.2. (1 pt) Which variable contains the size (ie. Number of elements) of the list?

1.3. (1 pt) Which variable contains the list of numbers?

1.4. (1 pt) What size (or type) are the variables a and p?

1.5. (3 pts) List the different addressing modes used in the code.

1.6. **(4 pts)** Modify the code as little as possible so that it sorts in descending order.

Part II: Programming (16 pts total)

1.
 - a. **(2 pts)** Give an abstract definition of a linked list
- Linked lists are the most basic self-referential structures. Linked lists allow you to have a chain of structs with related data.

a **linked list** is a data structure consisting of a group of nodes which together represent a sequence
- b. **(6 pts)** Write the algorithm of a function that inserts an element in a linked list of integers using FIFO.

```
void enqueue(char queue[], char x)
{
    if(REAR < MAX_SIZE - 1)
    {
        REAR++;
        queue[REAR] = x;
        QUEUESIZE++;
        if(FRONT == -1)
            FRONT++;
    }
    else
        cout<<"Queue Overflow";
}
```

2. **(8 pts)** Design a detailed algorithm to sort a given data of integers into ascending order using a bubble sort algorithm. Write a function that implements this algorithm in a programming language of your choice.

```
void bubble_sort(list[])
{
    int i, j, temp;
    for(i = 0; i < n; i++){
        for(j = n-1; j > i; j--){
            if(list[j] < list[j-1]){
                temp = list[j];
                list[j] = list[j-1];
                list[j-1] = temp;
            } //swap adjacent elements
        } //end of inner loop
    } //end of outer loop
} //end of bubble_sort
```

Part III: Software Engineering (14 pts total)

1. **(3 pts)** Compare and contrast procedural and object-oriented programming languages.
2. **(3 pts)** What is the relationship between abstract data types and classes?
3. **(4 pts)** Give three phases of your choice among the phases of the software development life cycle and give, for each phase, its role in the software development process. A maximum of one sentence is accepted for each phase.

4. **(4 pts)** Choose one of the following topic (a, b or c) and explain very briefly.
- a. Waterfall, Spiral, and Prototyping models. For what type of development projects are each of these software development models used?
 - b. Software metrics. Discuss size, complexity, and quality metrics.
 - c. Structural and functional software testing methods. Discuss the various structural and functional testing techniques, which are commonly used in software development projects.

Give chosen topic here (a, b or c): _____

Part 4: Database Systems (15 pts total)

1. FIFA would like to keep track of information about events of the World cup 2002 in Korea/Japan. The following are definitions of some of the tables identified to organize the database for FIFA. Use this information to answer subsequent questions.

Table Definitions

Player (Id, Name, Specialization, Country)

Foreign key – Country references History

Played (Id, Match, Goals, Cards, Fouls, Offsites)

Foreign key – Id references Player, Match references Games

Games (Match, Stadium, Date)

Foreign key – Stadium references Places

Places (Stadium, Host)

History (Country, Population, Worldcup, Won)

Dictionary describing attribute names used in the tables

Id- Official id given by FIFA to each player

Name – Family name of a player

Specialization – Roll of the player (goalkeeper, defender, midfielder, striker...)

Country – Country the player is playing for

Date – Date on witch game is played

Match – Official match number given by FIFA to each game

Stadium – Name of stadium where game is played

Host – Host country (Japan or Korea)

Goals – Number of goals scored by a player

Cards – Number of cards seen by a player in a particular match

Fouls – Number of fouls committed by a player in a particular match

Offsites – Number of offsites counted on a player in a particular match

Population – Population of a country

Worldcup – Number of time the country has participated in previous world cups

Won – Number of times a country has won the world cups before

Using the above information, write an SQL statement that performs the following:

(2 pts) Create a table with name Dangerous-Players and store Id, Name, Country, Match and Fouls for players who committed more than 3 fouls in a single game.

(3 pts) Display Match, Stadium, Date and Host for matches in which more than four goals are scored.

(3 pts) Display Name, Id and Country of a player who came from a country with good world cup history (won world cup at least ones) and who has never played in any game in this world cup (stayed on a reserve bench).

2. **(4 pts)** Normalization is one of the important aspects of relational database design. Discuss the first two normal forms (1NF, 2NF). Elaborate your answer using Examples.
3. **(3 pts)** Concurrency is one of the major problem areas that the database designer has to deal with in the shared database environments. Describe very briefly any two of such concurrency problems and discuss methods used to overcome the problems.

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Part 5: Operating system (14 pts total)

- 1) **(2 pts)** What are the two major purposes of an operating system?
- 2) **(3 pts)** Describe the three process states and the functions involved in the transition of processes from one state to another. Use a diagram to demonstrate your answer.
- 3) **(3 pts)** Selection of the value of the quantum time in CPU scheduling is one important aspect of the algorithms. Discuss the problems associated with the selection of a very small quantum value and a very large one?

- 4) **(6 pts)** Write a program that runs on UNIX (i.e. that uses UNIX system calls) and creates two processes. The first process should write PING, then the second should write PONG, then the first should write PING, then the second should write PONG, etc. indefinitely. You can use any communication and synchronization method available on UNIX and any programming language.

Part 6: Networking (15 pts total)

- 1) **(3 pts)** Choose two layers out of the seven layers of the ISO/OSI Networking model. Describe in a few words the main role (function) of each of the two layers that you have chosen (1 small sentence for each layer)

Layer Name	Description

- 2) **(2 pts)** Give at least two popular physical media used for computer networking. Compare briefly their characteristics for communication.

- 3) **(2.5 pts)** Explain briefly what Carrier Sense Multiple Access/ Collision Detection (CSMA/CD) is, where it is used and what are its two main concepts.

4) **(2.5 pts)** Given a base IP address 195.201.121.0 and using the first three bits of the host address part of the IP address as a sub network prefix:

a. What is the standard class of this IP address?

b. What will be the subnet mask?

c. What will be the number of usable subnets?

d. What will be the number of unique host addresses supported in each subnet?

e. What will be the first available IP address that can be assigned to a PC connected in the third usable subnet?

5) **(5 pts)** Write the letter of the descriptions under column B that best matches phrases under column A in the space provided.

Answer	Column A	Column B
H	1. Circuit Switching	A. Ensures that external communications confirm to the local network security policy.
F	2. DNS	B. Responsible for assigning Internet addresses to users.
J	3. Gateway	C. Physical layer protocol of X.25.
I	4. Routers	D. One of the ISDN channel services provided at a single interface.
A	5. Fire Wall	E. Packet switching technology that involves fixed length data unit.
	6. FDM or TDM	F. Converts numerical IP address in to readable text form.
	7. RS-232	G. Allows several transmission sources to share a larger transmission capacity.
	8. Cell Relay	H. Lines are dedicated for the duration of the connection.
	9. 23B+D	I. Make use of RIP to transfer data packets across multiple networks.
B	10. ISP	J. Performs translation between two dissimilar networks.

General questions on ICT in Ethiopia and abroad (15 pts total)

Instruction: Attempt any two of the following five questions. **If you attempt more than two none will be accepted.**

Each question is given 7.5 points as long as you follow the above instructions.

1. IBM mainframe computers used to be called a closed/proprietary system: What does it mean?
2. Do you see advantages/drawbacks in liberalizing the telecom sector in the Ethiopian context? Justify your answer.

3. List three major obstacles against the development of ICT in Ethiopia. Argument your answer.
4. It is nowadays very commonly said that there is convergence of information technologies and communication technologies. What does this statement mean? Give examples to justify your answers.
5. State and explain two ethical rules an IT professional must follow in Ethiopia.